Use-Case Flow Specification

Use Case: Process Multi-Vendor Customer Order

Actors:

- **Primary Actor:** Storefront Platform (Shopify/WooCommerce, via webhook)
- **Secondary Actors:** Supplier Partner (receives PO, provides inventory data), E-Store Owner (receives status notifications)

Related Use Cases:

- Includes: *Verify Supplier Inventory*
- Extends (conditionally): *Handle Backorder*

PRECONDITIONS

1. The webhook endpoint is authenticated and reachable, and the store's webhook secret has been validated.
2. A SKU-to-supplier catalog mapping exists and is up to date for the store.
3. Supplier API credentials for all relevant suppliers are valid and active.

POSTCONDITIONS (ON SUCCESS)

1. One purchase order has been generated and dispatched per supplier represented in the original order.
2. Each dispatched PO carries the correct line items and the customer's delivery address.
3. The parent order's status is updated to "Processing" (or "Partially Backordered" if the alternate flow was triggered).
4. The E-Store Owner's dashboard reflects the current split/dispatch status of the order.

MAIN SUCCESS SCENARIO

1. The Storefront Platform sends an order webhook to the system.
2. The system acknowledges receipt with an HTTP 200 response within 200 ms (per NFR-001).
3. The system parses the webhook payload, extracting line items, customer details, and the delivery address.
4. The system maps each ordered SKU to its owning Supplier Partner using the catalog mapping.
5. The system groups line items by supplier, producing one sub-order per supplier (**Split Multi-Vendor Order**).
6. For each supplier group, the system performs a real-time inventory check against the supplier's API (**«include» Verify Supplier Inventory**).
7. All queried SKUs are confirmed in stock.

8. The system generates and dispatches a purchase order to each Supplier Partner, including the customer's delivery address and order reference.
9. The system updates the parent order status to "Processing" and records the generated PO IDs against it.
10. The system notifies the E-Store Owner's dashboard that the order has been split and successfully dispatched.

ALTERNATE FLOW: SUPPLIER SKU OUT OF STOCK

Branches from Main Success Scenario at Step 7.

7a. The inventory check for one supplier group returns insufficient stock for one or more SKUs.

7b. The system triggers **Handle Backorder** (extension point: "stock insufficient" on Verify Supplier Inventory).

7c. The system withholds PO dispatch for only the affected supplier group; all other, in-stock supplier groups continue through Steps 8–10 unaffected.

7d. The system notifies the E-Store Owner of the specific backordered items and the affected supplier (per FR-005).

7e. The system marks the affected order segment as "Backordered" while unaffected segments proceed to "Processing."

7f. The use case resumes at Step 9 for the remaining, successfully dispatched groups.

Resulting Postcondition: The parent order ends in a mixed state — some line items "Processing" (PO dispatched) and others "Backordered" (dispatch withheld) — with the E-Store Owner notified of the split outcome.